بسم الله الرحمن الرحيم

Numbering System

Decimal

$10 = 2^{3+1}$

0,1,2,3,4,5,6,7,8,9

10 = عدد الرموز

Radix a base

Binary

Computer

0,1

2 = عدد الرموز

Radix a base

Hexadecimal

$6 + 10 = 16 = 2^4$

0,1,2,...,9,A,B,C,D,E,F

10/11/12/13/14/15

16 = عدد الرموز

Radix a base

Octal

$8 = 2^3$

0,1,2,3,4,5,6,7

8 = عدد الرموز

Radix a base

Radix-R

$R = 2^?$

0,1,2,3,...,(R-1)

R = عدد الرموز

Radix a base

$(1709)_{10}$ — Weights: $*1 = 10^0$, $*10 = 10^1$, $*100 = 10^2$, $*1000 = 10^3$

1709.123 — $10^3\ 10^2\ 10^1\ 10^0\ 10^{-1}\ 10^{-2}\ 10^{-3}$

$(1709)_{10}$: $*1 = 9*1 = 9$; $*10 = 0*10 = 0$; $*100 = 7*100 = 700$; $*1000 = 1*1000 = 1000$ → $(1709)_{10}$

$(1101)_2$ — Weights: $*2^0$, $*2^1$, $*2^2$, 2^3

$(1101)_2$: $*1 = 1*1 = 1$; $*2 = 0*2 = 0$; $*4 = 1*4 = 4$; $*8 = 1*8 = 8$ → $(13)_{10}$

$(1F2C)_{16}$ — Weights: $*0$ (16), $*16^1$, $*16^2$, $*16^3$

$(1F2C)_{16}$: $*1 = 12*1 = 12$; $*16 = 2*16 = 32$; $*256 = 15*256 = 2560$; $*4096 = 1*4096 = 4096$ → $(7980)_{10}$

376 — Weights: $*8^0$, $*8^1$, $*8^2$

376: $*1 = 6*1 = 6$; $*8 = 7*8 = 56$; $*64 = 3*64 = 192$ → $(254)_{10}$

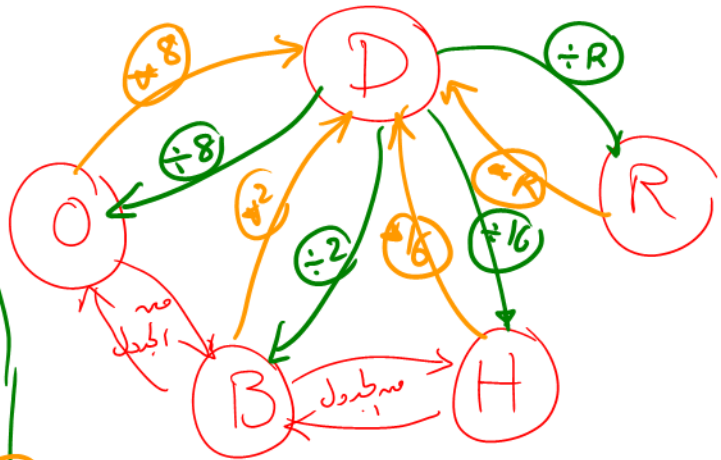
$(132)_R$ — Weights: $*R^0$, $*R^1$, $*R^2$

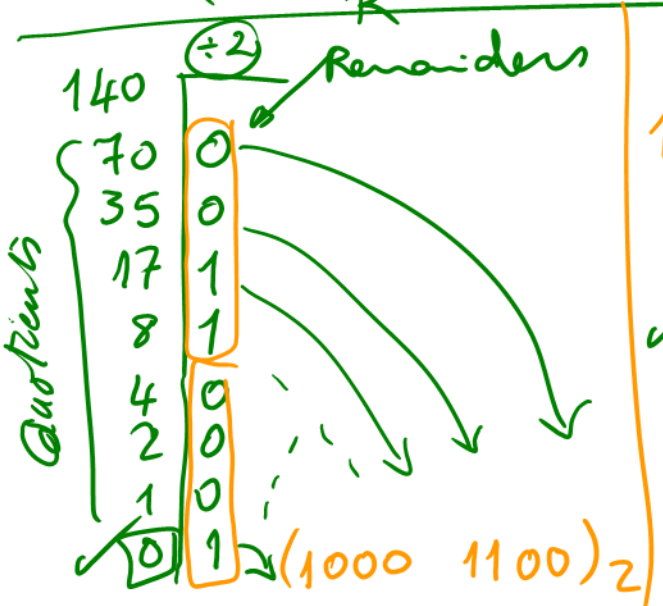
$(140)_{10} = (1000\ 1100)_2$

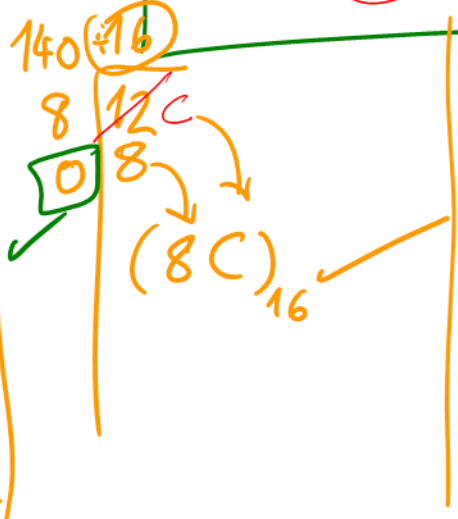
$= (8C)_{16}$

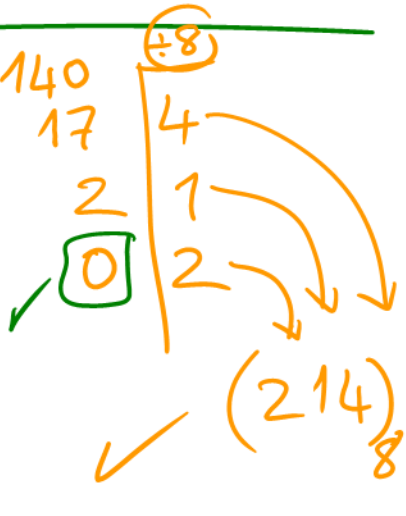
$= (214)_8$

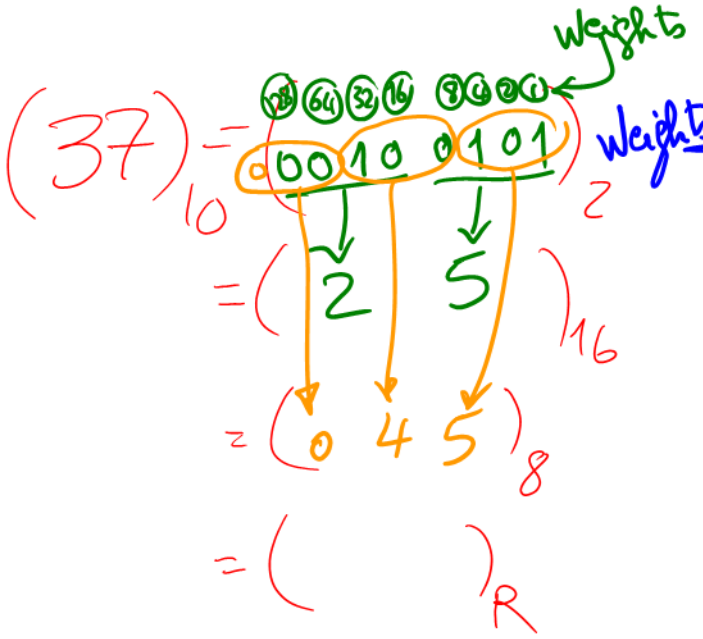
$= (\quad)_R$











H	O	B	D
①	⑧ ①	⑧ ④ ② ①	⑩ ①
0	0 0	0 0 0 0	0 0
1	0 1	0 0 0 1	0 1
2	0 2	0 0 1 0	0 2
3	0 3	0 0 1 1	0 3
4	0 4	0 1 0 0	0 4
5	0 5	0 1 0 1	0 5
6	0 6	0 1 1 0	0 6
7	0 7	0 1 1 1	0 7
8	1 0	1 0 0 0	0 8
9	1 1	1 0 0 1	0 9
A	1 2	1 0 1 0	1 0
B	1 3	1 0 1 1	1 1
C	1 4	1 1 0 0	1 2
D	1 5	1 1 0 1	1 3
E	1 6	1 1 1 0	1 4
F	1 7	1 1 1 1	1 5

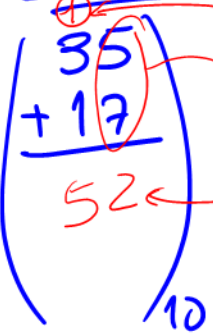
37
-32
5
-4
1
-1
0 ✓

(25)10
*1 = 5
*16 ⇒ 32+
37 ✓

(45)8
*1 = 5
*8 = 32+
37 ✓

Arithmetics

1 Additrei



Rules

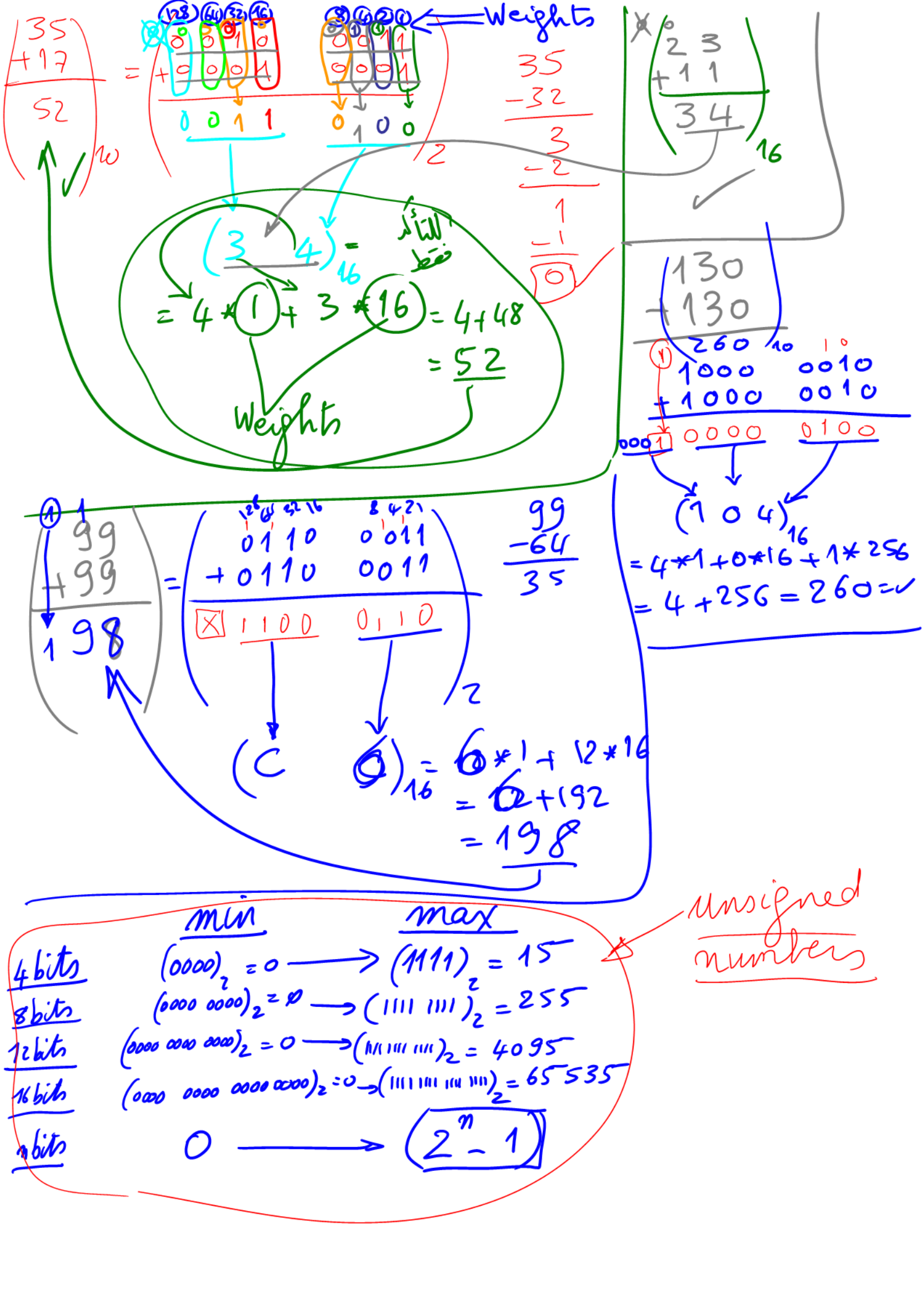
0 + 0	0 + 1	1 + 0	1 + 1
0 0	0 1	0 1	1 0

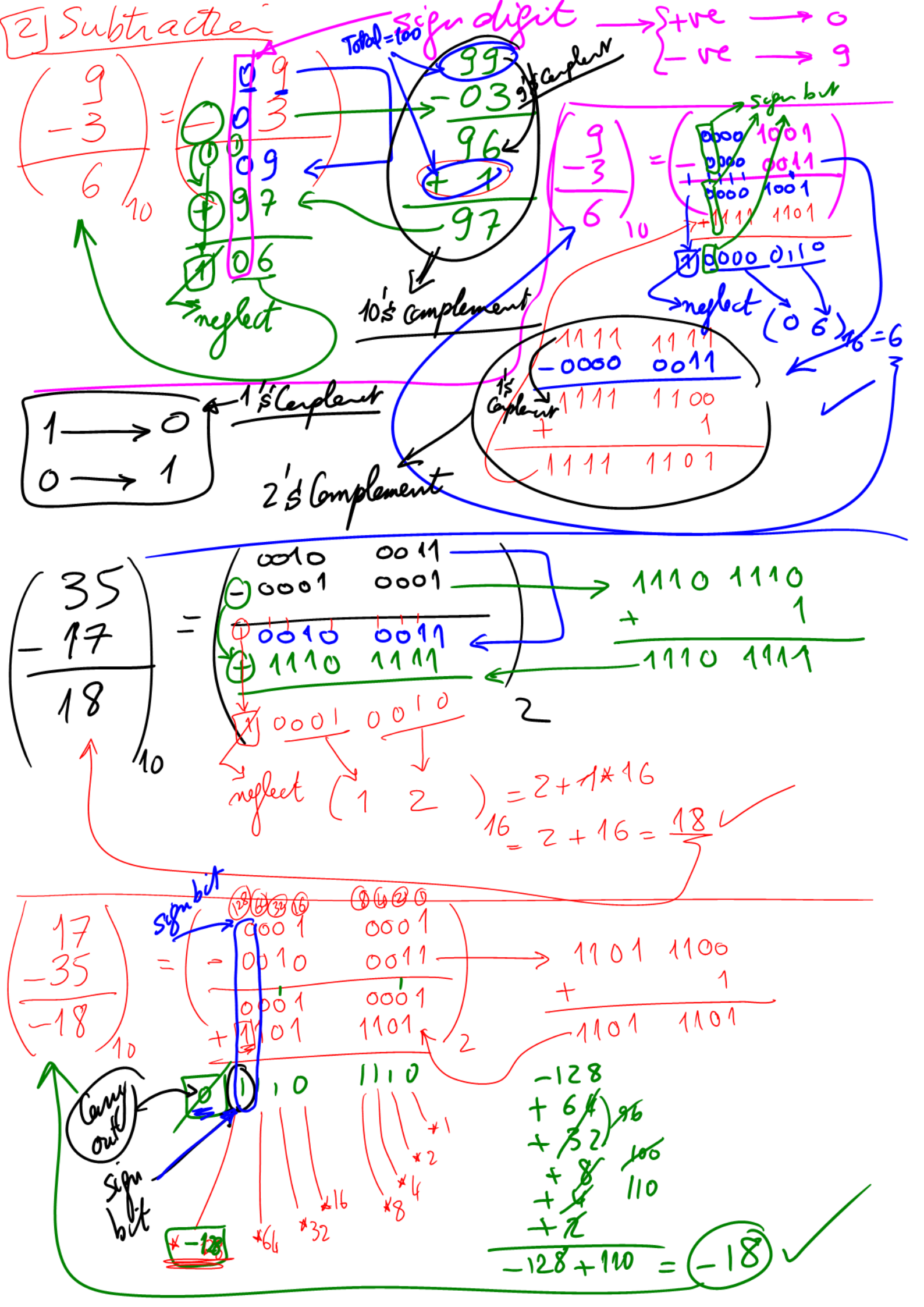
(2)10 = (10)2

Carry Sum

1 + 1 + 1 = 11

(3)10 = (11)2


35 + 17 = 52
Weights
35 - 32 = 3
3 - 2 = 1
1 - 1 = 0
(3 4)16 = 4*1 + 3*16 = 4+48 = 52
Weights
للتأكد فقط
2 3 + 1 1 = 34 16
130 + 130 = 260 10
1000 0010 + 1000 0010 = 0100
(1 0 4)16 = 4*1 + 0*16 + 1*256 = 4 + 256 = 260 ✓
99 + 99 = 198
99 - 64 = 35
0110 0011 + 0110 0011 = 1100 0110
(C 6)16 = 6*1 + 12*16 = 6 + 192 = 198
unsigned numbers
4 bits: min (0000)2 = 0 → max (1111)2 = 15
8 bits: min (0000 0000)2 = 0 → max (1111 1111)2 = 255
12 bits: min (0000 0000 0000)2 = 0 → max (1111 1111 1111)2 = 4095
16 bits: min (0000 0000 0000 0000)2 = 0 → max (1111 1111 1111 1111)2 = 65535
n bits: 0 → 2^n - 1


2] Subtraction
sign digit
+ve → 0
-ve → 9
Total = 100
99
- 03
96
+ 1
97
1's Complement
10's complement
neglect
06
sign bit
1 → 0
0 → 1
1's Complement
2's Complement
0000 1001
- 0000 0011
0000 1001
+ 1111 1101
0000 0110
neglect
(0 6)16 = 6
1111 1111
- 0000 0011
1111 1100
+ 1
1111 1101
35
- 17
18
0010 0011
- 0001 0001
0010 0011
+ 1110 1111
0001 0010
neglect (1 2)16 = 2 + 1*16 = 2 + 16 = 18
1110 1110
+ 1
1110 1111
17
- 35
- 18
sign bit
0001 0001
- 0010 0011
0001 0001
+ 1101 1101
1 0 1110
Carry out
sign bit
*-128
*64
*32
*16
*8
*4
*2
*1
1101 1100
+ 1
1101 1101
-128
+ 64
+ 32
+ 8
+ 4
+ 2
-128 + 110 = -18

	min	max
4 bits	$-2^{4-1} = -8$	$+2^{4-1} - 1 = 7$
8 bits	$-2^{8-1} = -128$	$+2^{8-1} - 1 = 127$
16 bits	$-2^{16-1} = -32268$	$+2^{16-1} - 1 = 32267$
n bits	-2^{n-1}	$+2^{n-1} - 1$

$-8 \rightarrow +7$

$-128 \rightarrow +127$

Signed Numbers

How to represent negative numbers?

1. Sign/Magnitude (0 +ve, 1 -ve)
2. 1's Complement
3. 2's Complement

3 Multiplication

35×17

$245 + 350 = 595_{10}$

$0010\ 0011 \times 0001 = 0010\ 0011$

$0010\ 0011 + 0010\ 0110 = 0010\ 0101\ 0011_2$

$(2\ 5\ 3)_{16} = 3*1 + 5*16 + 2*256$

$= 3 + 80 + 512$

$= 595$

جدول الضرب

0	0	1	1
*0	*1	*0	*1
0	0	0	1

4 Division

$\frac{35}{7} = 5\ \frac{0}{7}$

$\frac{37}{7} = 5\ \frac{2}{7}$

$\frac{11}{4} = 2\ \frac{3}{4}$

Quotient

Remainder

Fixed point format

$37000000 \div 7$

-35

20

-14

60

-56

40

-35

5

Remainder

5.2857

Quotient

Truncation error

float → 4 bytes = 32 bits
double → 8 bytes = 64 bits
} Floating point formats.

$e = 1.6 * 10^{-19}$ cb = 0.0000000000000000016

$(-3.52 * 10^{4})_{10} \neq (-1.1101001 * 2^{x})$

1000 0000

biased exponent

127 + 4

Sign

mantissa

exponent

32 | 1 | 8 | 23

1 | 1000 0000 | 1110 1001 0100 - - - 0000

IEEE 32-bit floating point format

3.5217 + 0.0015

3.5217
+0.0015
3.5232

fractional point

5.3
*0.2
1.06

0.4 * 2 = 0.8
0.8 * 2 = 1.6
0.6 * 2 = 1.2

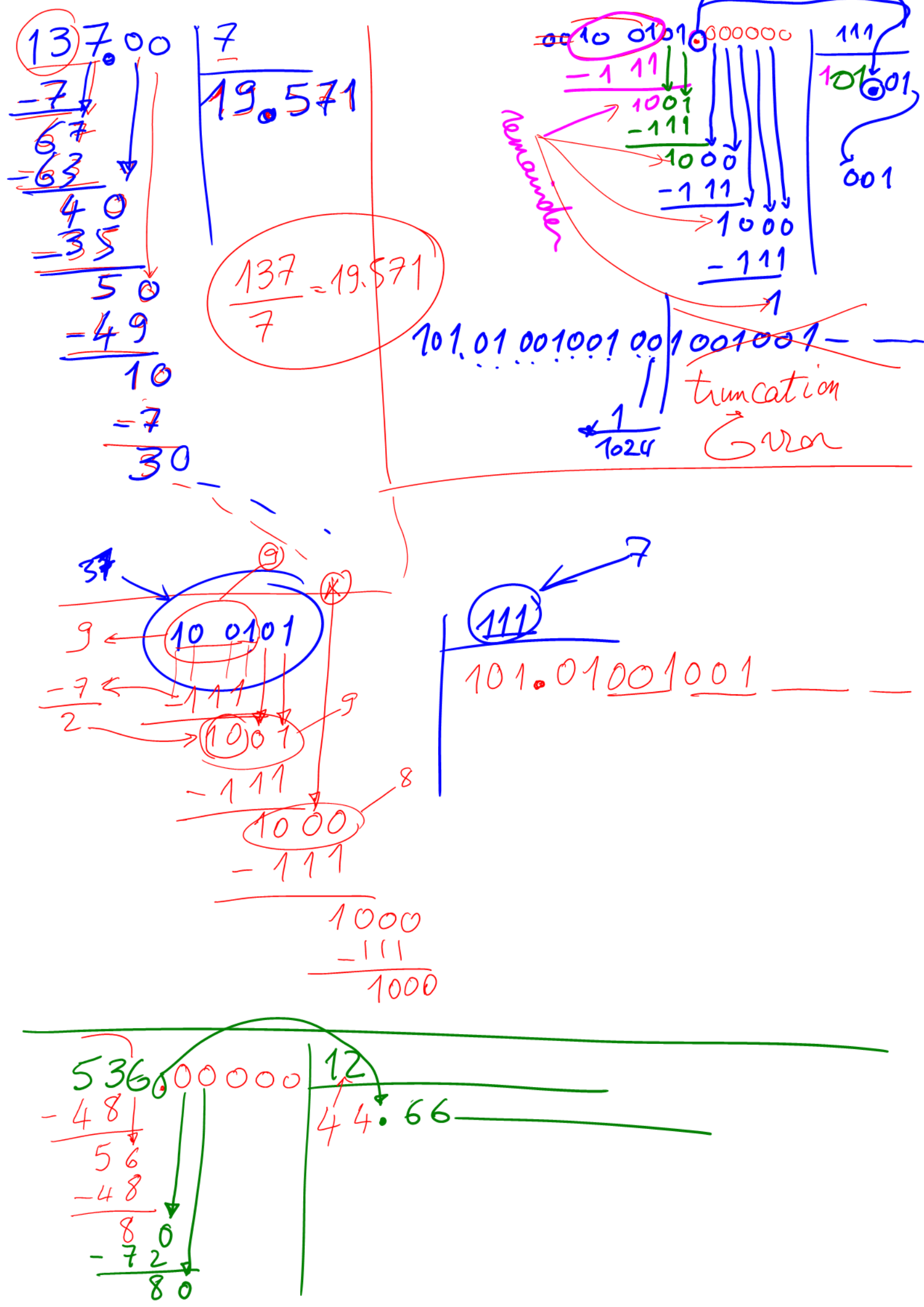
0.3 * 2 = 0.6
0.6 * 2 = 1.2
0.2 * 2 = 0.4
0.4 * 2 = 0.8
0.8 * 2 = 1.6
0.6 * 2 = 1.2
0.2 * 2 = 0.4

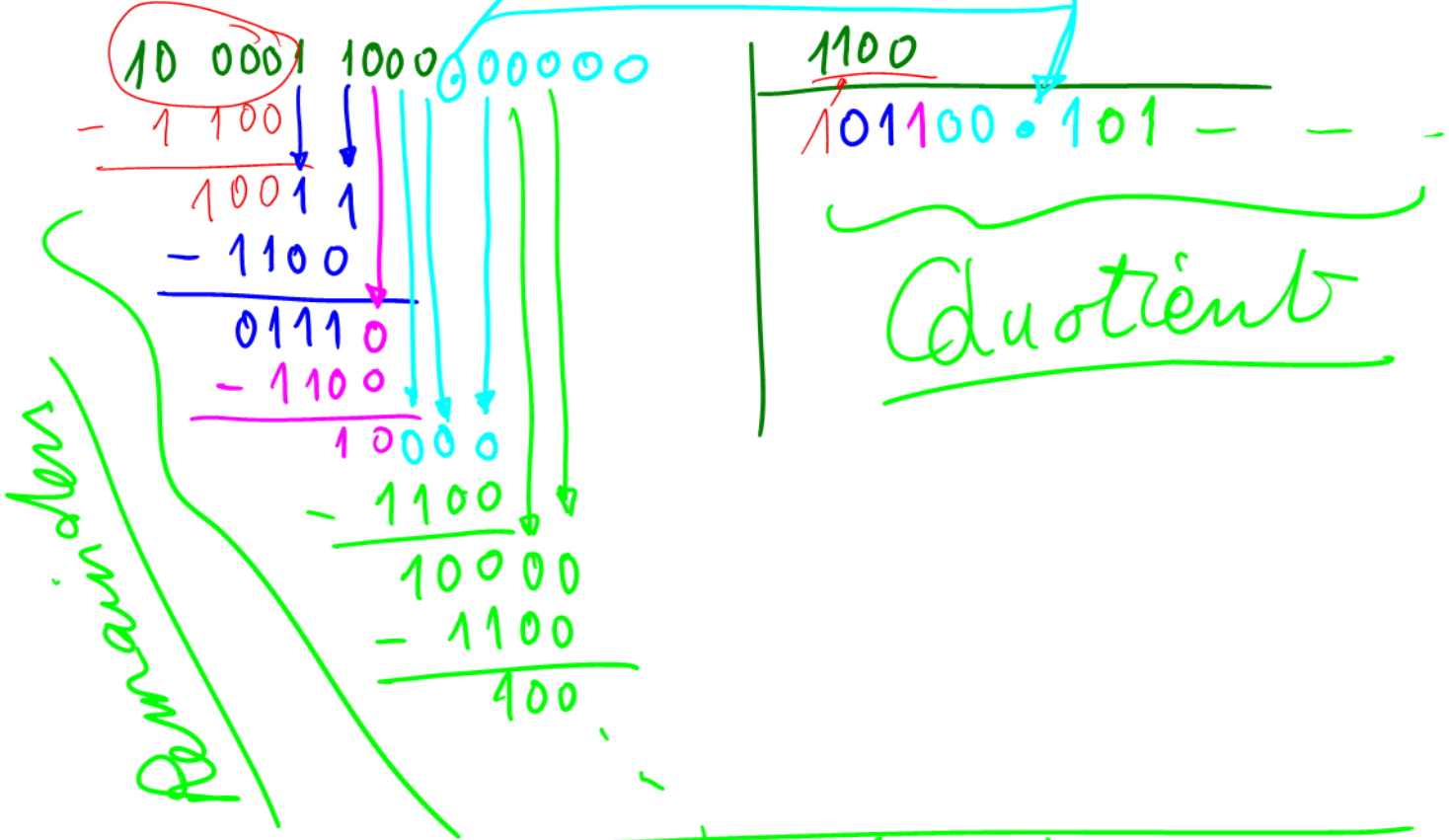
0101.0100110011...

0000.

0.1%

emca


137.00
7
19.571
137/7 = 19.571
10 01 01
111
101.01
001
remainder
101.01 001001 00
1001001
truncation
Error
1/1024
37
9
10 0101
111
7
101.01001001
1001
1000
1000
111
1000
536.000000
12
44.66
48
56
48
80
72
80



Binary Coded Decimals (BCD)

$(4)_{10} = (0000\ 0000\ 0100)_2 = (0000\ 0000\ 0100)_{BCD}$

$(41)_{10} = (0000\ 0010\ 1001)_2 = (0000\ 0100\ 0001)_{BCD}$

$(417)_{10} = (0001\ 1010\ 0001)_2 = (0100\ 0001\ 0111)_{BCD}$



BCD Arithmetics

1 Addition

BCD → 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Hexadecimal → 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

16 = عدد

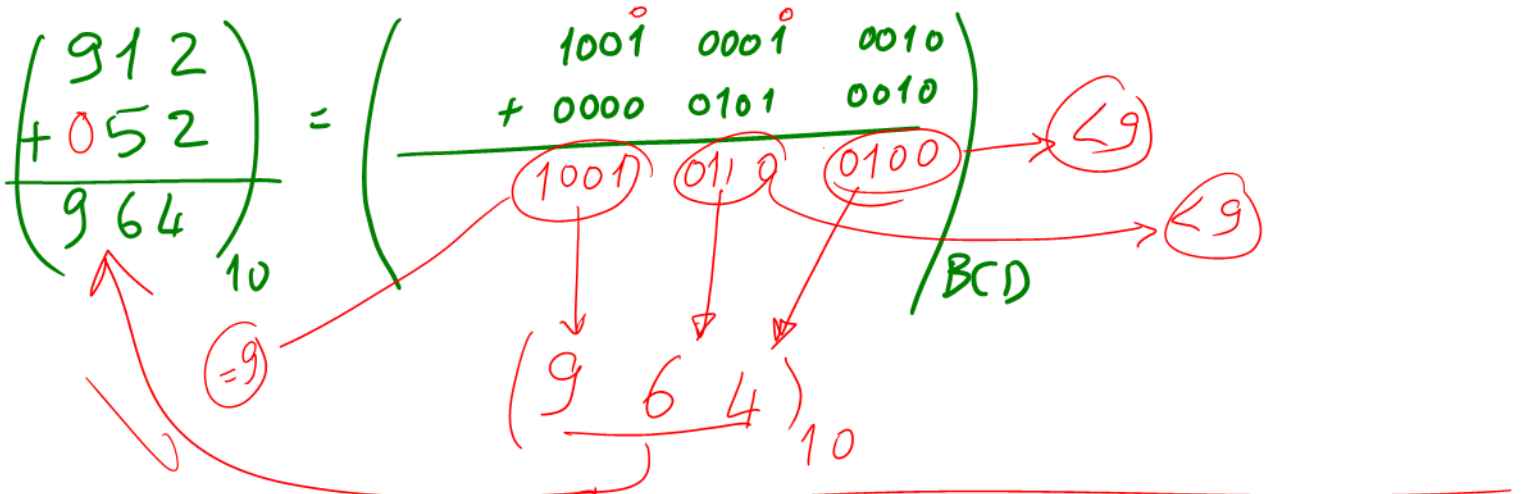
$$\begin{array}{r} 35 \\ +17 \\ \hline 52 \end{array}_{10} = \left(\begin{array}{r} 0011 \\ +\ 0001 \\ \hline 0101 \end{array} \quad \begin{array}{r} 0101 \\ 0111 \\ \hline 1100 \\ +0110 \\ \hline 10010 \end{array} \right)_{BCD}$$

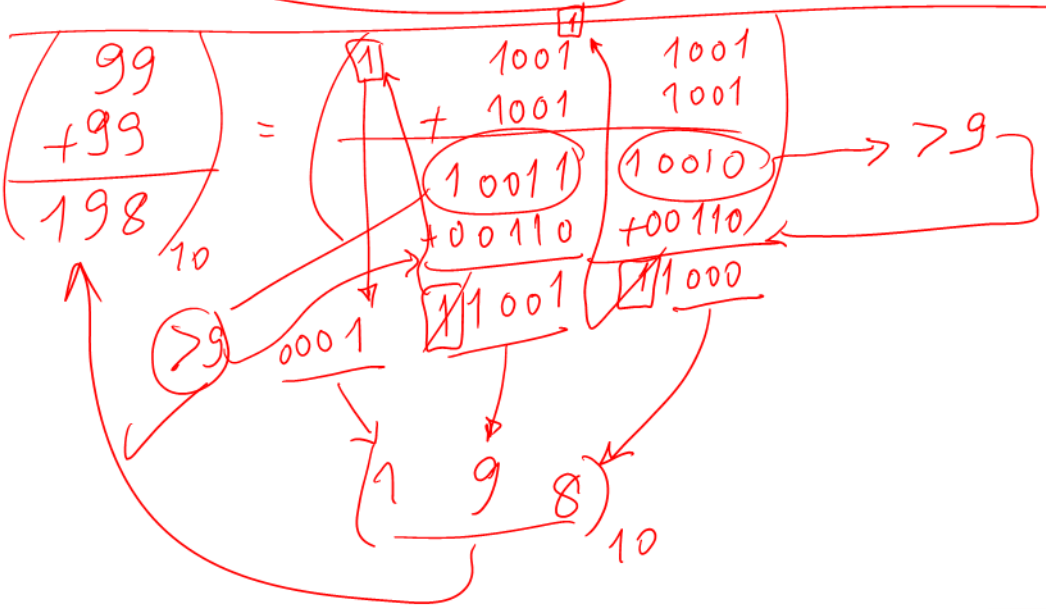
$(5\ \ 2)_{10}$

Rules:

If result > 9 ⇒ Add 6

If result ≤ 9 ⇒ Add 0

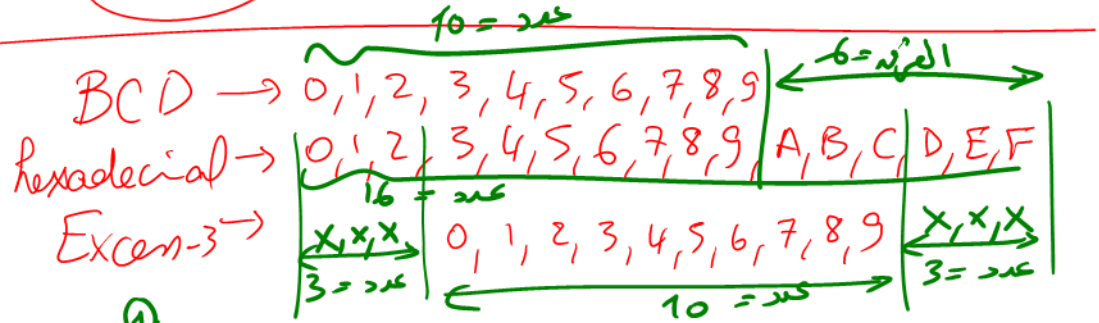


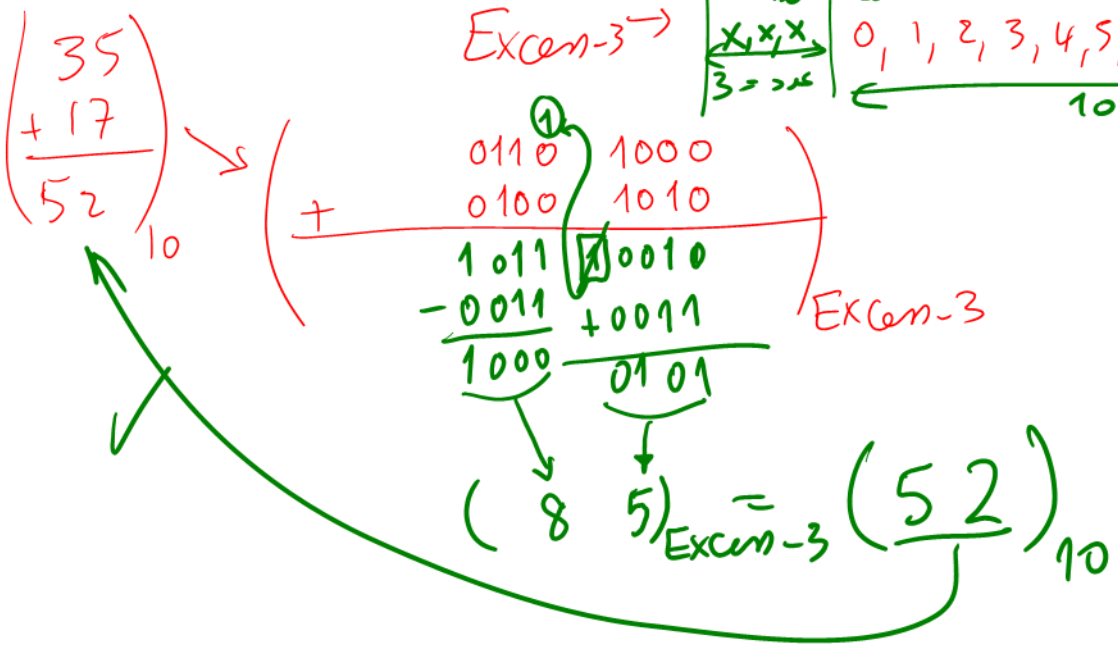


2 Subtraction 3 Multiplication 4 Division

HW

Excess-3





$$\begin{array}{r} 99 \\ +99 \\ \hline 198 \end{array}_{10} = \left(\begin{array}{r} 1100\ 1100 \\ +\ 1100\ 1100 \\ \hline 0001\ 1001\ 1000 \\ +0011\ +0011\ +0011 \\ \hline 0100\ 1100\ 1011 \end{array} \right)_{\text{Excess-3}}$$

$$(4\ C\ B)_{\text{Excess-3}} = (198)_{10}$$

✓

2] Subtracție 3] Multiplicație 4] Diviziune

HW

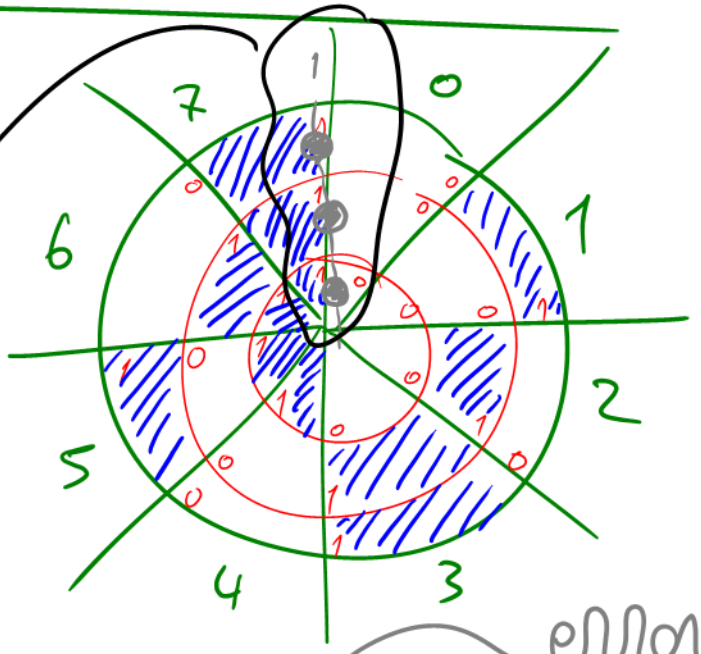
Gray Code

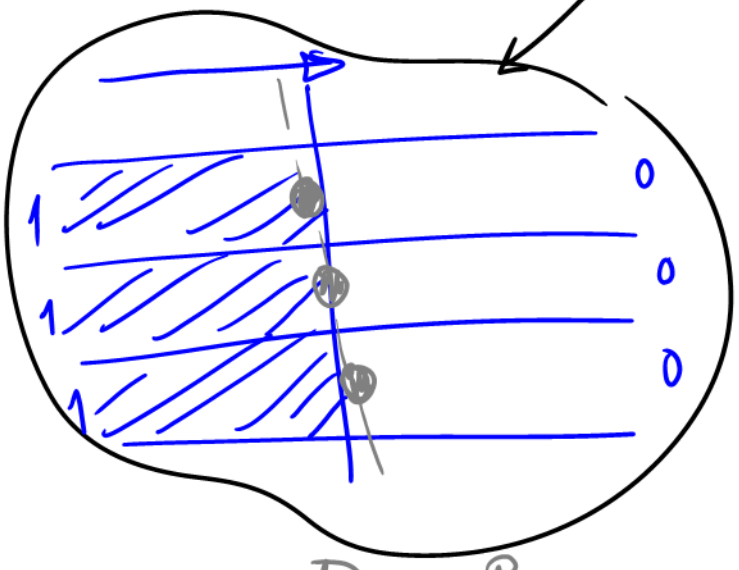
0 → 7

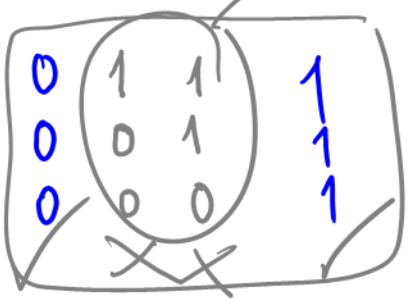
000 → 111

□ ⇒ 0

▨ ⇒ 1





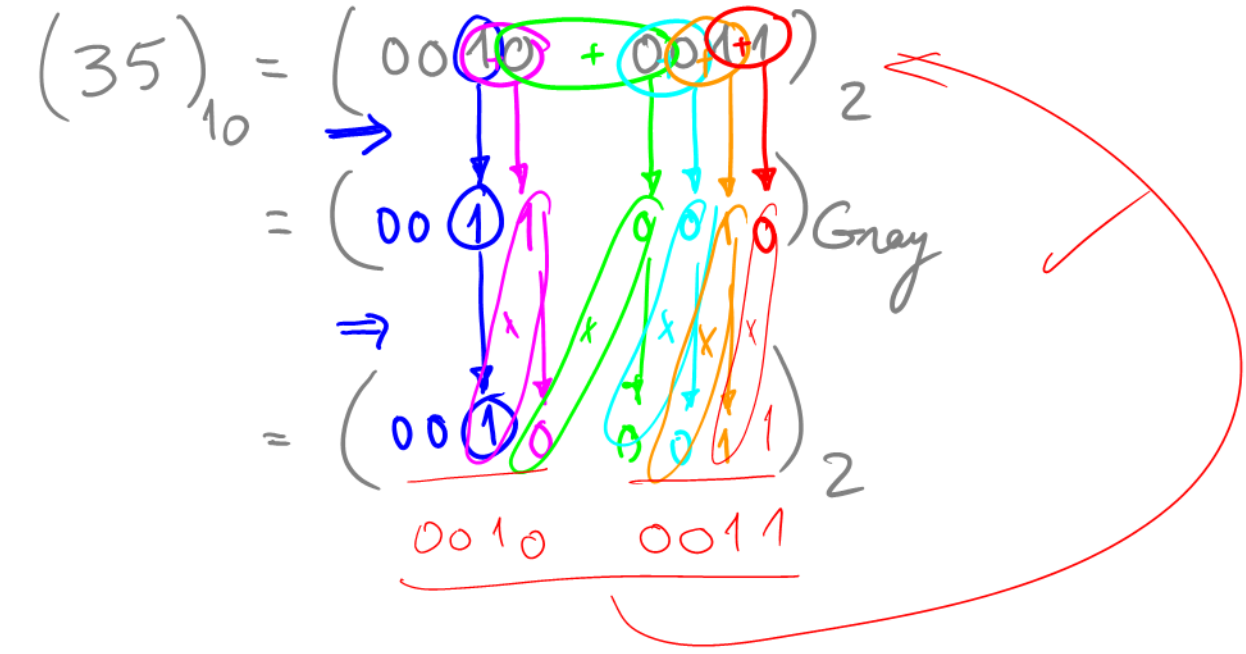


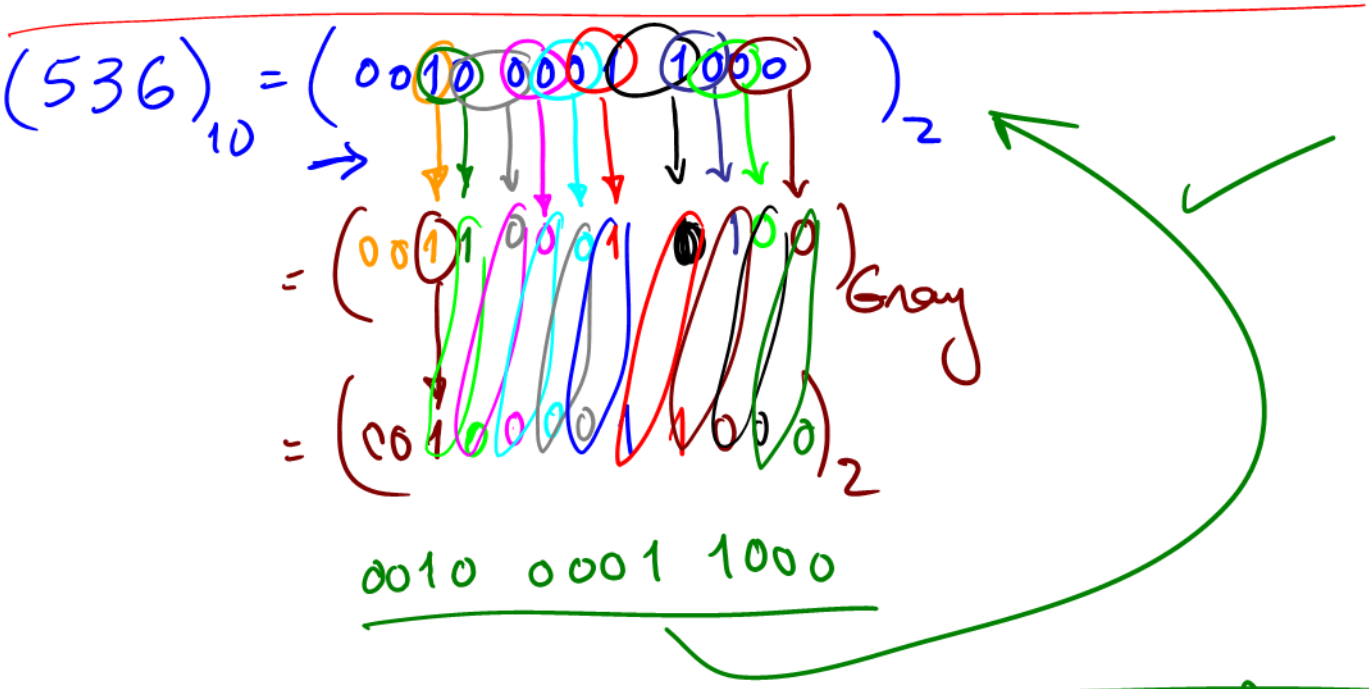
error!!!

D	B	G
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100
	000	000

Only 1 change per step

Gray Code is not an Arithmetic Code



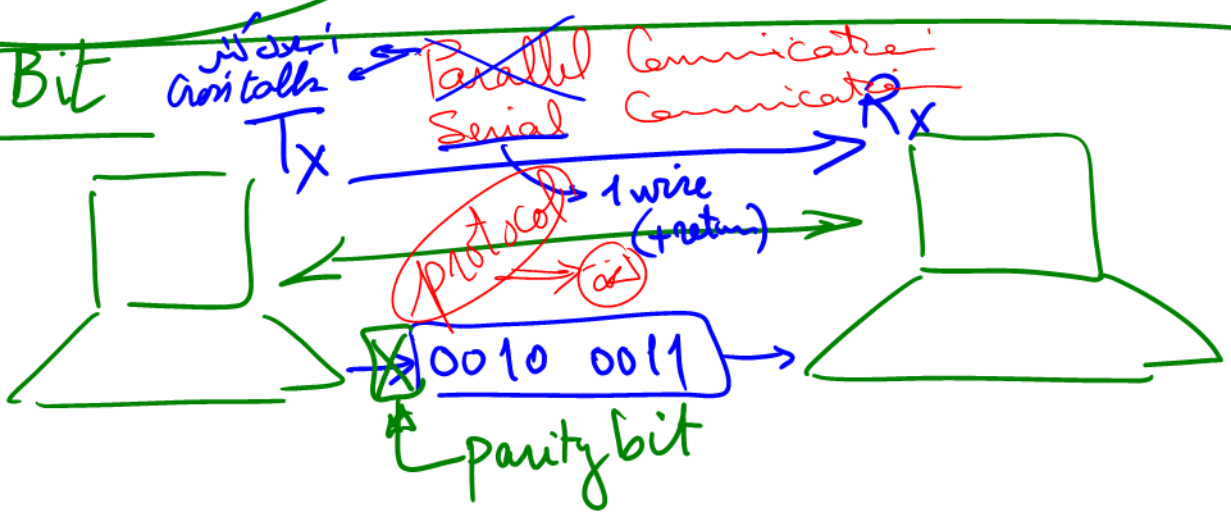


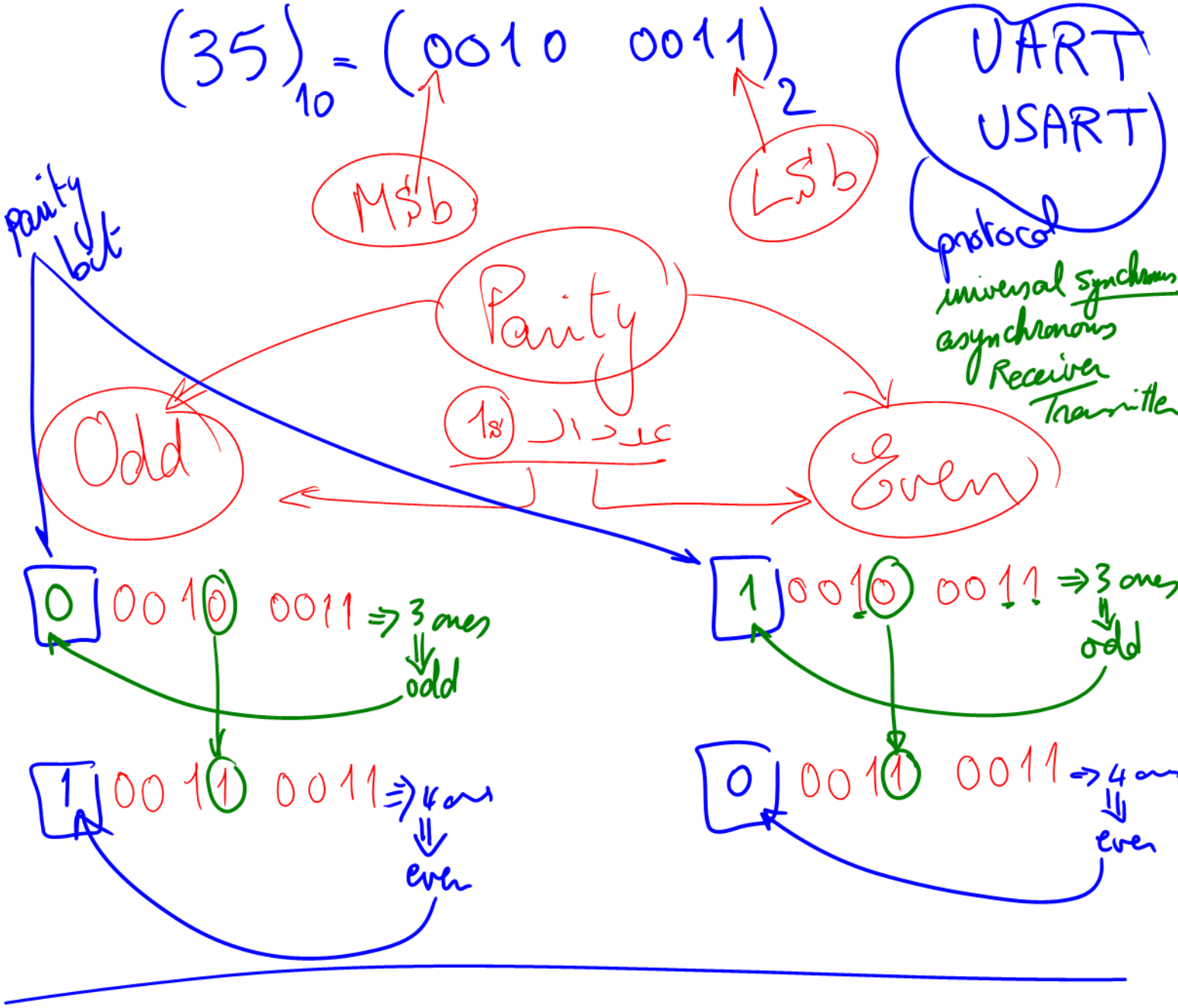
ASCII Code

American Standard Code for Information Interchange

This is not an Arithmetic Code

Parity Bit




(35)10 = (0010 0011)2
MSb
LSb
UART
USART
protocol
universal synchronous asynchronous Receiver Transmitter
Parity bit
Parity
عدد ال 1s
Odd
Even
0 0010 0011 ⇒ 3 ones ⇓ odd
1 0010 0011 ⇒ 3 ones ⇓ odd
1 0010 0011 ⇒ 4 ones ⇓ even
0 0010 0011 ⇒ 4 ones ⇓ even